

DiffCSP: Finding Browser Bugs in Content Security Policy Enforcement through Differential Testing

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The Content Security Policy (CSP) is one of the de facto security mechanisms that mitigate web threats. Many websites have been deploying CSPs mainly to mitigate cross-site scripting (XSS) attacks by instructing client browsers to constrain JavaScript (JS) execution. However, a browser bug in CSP enforcement enables an adversary to bypass a deployed CSP, posing a security threat. As the CSP specification evolves, CSP becomes more complicated in supporting an increasing number of directives, which brings additional complexity to implementing correct enforcement behaviors. Unfortunately, the finding of CSP enforcement bugs in a systematic way has been largely understudied.

In this paper, we propose DiffCSP, the first differential testing framework to find CSP enforcement bugs involving JS execution. DiffCSP generates CSPs and a comprehensive set of HTML instances that exhibit all known ways of executing JS snippets. DiffCSP then executes each HTML instance for each generated policy across different browsers, thereby collecting inconsistent execution results. To analyze a large volume of the execution results, we leverage a decision tree and identify common causes of the observed inconsistencies. We demonstrate the efficacy of DiffCSP by finding 29 security bugs and eight functional bugs. We also show that three bugs are due to unclear descriptions of the CSP specification. We further identify the common root causes of CSP enforcement bugs, such as incorrect CSP inheritance and hash handling. We confirm the risky trend of client browsers deriving completely different interpretations from the same CSPs, which

raises security concerns. Our study demonstrates the effectiveness of DiffCSP for identifying CSP enforcement bugs, and our findings have contributed to patching 12 security bugs in major browsers, including Chrome and Safari.

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